



UNIVERSITY OF MORATUWA
Faculty of Information Technology

B.Sc. in Information Technology & Management
Semester 2

IN1130 – STRUCTURED PROGRAMMING II

Submission Deadline: 04-02-2026

Question 01

(a) Choose **ONE real-world system** from the list below **OR** propose your own.

- Library Management System
- Clinic / Hospital Patient Record System
- Supermarket Billing System
- Movie Ticket Booking System
- Employee Payroll System
- Hotel Room Reservation System

(b) Write a short description answering,

- What problem does this system solve?
- Who uses this system?
- What type of information must be stored?

[15 Marks]

(c) Identify and write:

- The **main entity** of the system
- At least **4 attributes** related to that entity

E.g.- Entity: Student

Attributes: name, index number, GPA, degree program

[10 Marks]

Continued....

(d) Based on your chosen system:

- Design a structure to represent the entity
- Clearly mention:
 - Attribute name
 - Data type
 - Reason for choosing that data type

[15 Marks]

(e) Answer the following:

- Will the system store **one record** or **multiple records**?
- Justify whether an **array of structures** is required.

[5 Marks]

(f) Write a C program that:

- Uses your designed structure
- Stores details of **at least 3 records**
- Accepts input from the user

[30 Marks]

(g) Your program must perform **at least TWO meaningful operations**, such as:

- Calculate totals or averages
- Search for a record
- Update a value
- Count records matching a condition

[10 Marks]

(h) Display the results in a **professional, real-world format**, such as:

- Bill
- Report
- ID card
- Summary sheet

[5 Marks]

Continued...

(i) Explain:

- Why you grouped the selected attributes together
- How using structures improved your solution

[8 Marks]

(j) Answer:

- What problems would occur if you used **only arrays and variables** instead of structures?

[2 Marks]